Status	Started	Completed
Finished	Wednesday, 11 November 2026, 12:57 PM	Wednesday, 11 November 2026, 1:00 PM

Duration	Grade
3 mins 17 secs	**20.00** out of 20.00 (**100**%)

Question 1

Flag question

Correct Mark 1.00 out of 1.00

Your research coordinator assigns the same topic to two different subagents 'for redundancy'. Both complete successfully with nearly identical findings. The synthesis agent now has doubled content to process, increasing context usage and latency. What decomposition principle should be applied instead?

a. Assign one subagent as primary and have it delegate to the second only on failure.

b. Partition the research scope so each subagent covers a distinct and non-overlapping slice. Redundancy increases cost without improving quality; orthogonal decomposition (by geography, time period, source type, etc.) provides broader coverage more efficiently. ✓

c. Reduce to a single subagent per topic to eliminate redundancy entirely.

d. Redundancy is valuable — keep both subagents for thoroughness.

The correct answer is: Partition the research scope so each subagent covers a distinct and non-overlapping slice. Redundancy increases cost without improving quality; orthogonal decomposition (by geography, time period, source type, etc.) provides broader coverage more efficiently.

Question 2

Flag question

Correct Mark 1.00 out of 1.00

Your coordinator agent selects subagents dynamically. For a query asking 'summarise this uploaded PDF', the coordinator invokes web_search, document_analysis, and synthesis. Which principle does this violate?

a. The coordinator must always invoke at least two subagents for quality assurance.

b. Model-driven dynamic selection: the coordinator should analyse query requirements and invoke only necessary subagents. A PDF summarisation requires only document_analysis — web_search adds latency and cost without value. ✓

c. The coordinator should never invoke synthesis directly.

d. PDF queries must always bypass the coordinator and go directly to document_analysis.

The correct answer is: Model-driven dynamic selection: the coordinator should analyse query requirements and invoke only necessary subagents. A PDF summarisation requires only document_analysis — web_search adds latency and cost without value.

Question 3

Flag question

Correct Mark 1.00 out of 1.00

You implement a PreToolUse hook to block calls to send_email until the agent has called verify_recipient. During testing you find the hook never fires for send_email. What is the most likely cause?

a. Hooks only fire after stop_reason is 'tool_use', not before.
b. PreToolUse hooks require the tool name to be in uppercase.
c. The send_email tool is not included in the agent's allowed_tools list, so it is never callable and the hook has no calls to intercept — the tool must be in allowed_tools for hooks to fire. ✓
d. PreToolUse hooks only work with MCP tools, not built-in tools.

The correct answer is: The send_email tool is not included in the agent's allowed_tools list, so it is never callable and the hook has no calls to intercept — the tool must be in allowed_tools for hooks to fire.

Question 4

Flag question

Correct Mark 1.00 out of 1.00

Your agent is configured with allowed_tools: ['get_customer', 'lookup_order', 'process_refund']. A new requirement needs the agent to also send confirmation emails. A developer adds send_email to allowed_tools. What principle should govern this decision?

a. Add all potentially useful tools — a broader tool set improves agent capability.
b. Consult the team before adding any new tool to allowed_tools.
c. Apply least-privilege tool assignment: add send_email only if this agent's role explicitly requires email sending. Consider whether a separate specialised subagent should handle email operations. Tool sets should be minimal and role-appropriate to maintain reliable tool selection. ✓
d. Tools should be added freely and removed later if they cause problems.

The correct answer is: Apply least-privilege tool assignment: add send_email only if this agent's role explicitly requires email sending. Consider whether a separate specialised subagent should handle email operations. Tool sets should be minimal and role-appropriate to maintain reliable tool selection.

Question 5

Flag question

Correct Mark 1.00 out of 1.00

Your agent for processing legal documents occasionally skips a mandatory document authentication step when clients say 'I've already authenticated this.' What is the most reliable fix?

a. Update the system prompt to clarify that authentication cannot be skipped.
b. Require users to re-authenticate every 30 minutes regardless of what they claim.
c. Implement a programmatic prerequisite gate: a PreToolUse hook blocks all document processing tools unless authenticate_document has been called and returned a confirmed authentication result in the current session. User assertions about prior authentication cannot override programmatic enforcement. ✓
d. Add an authentication confirmation step at the beginning of every session.

The correct answer is: Implement a programmatic prerequisite gate: a PreToolUse hook blocks all document processing tools unless authenticate_document has been called and returned a confirmed authentication result in the current session. User assertions about prior authentication cannot override programmatic enforcement.

Question 6

Flag question

Correct Mark 1.00 out of 1.00

Your coordinator agent is responsible for a 6-hour research task. At hour 4, a developer checks in and finds the coordinator is still on subtask 2 of 8. Investigation shows the coordinator is spending excessive time waiting for each subagent to complete before dispatching the next. What is the fix?

a. Reduce the number of subtasks from 8 to 4 to cut the total work.
b. Identify which subtasks are independent and dispatch them as parallel Task calls in single coordinator responses. Only tasks with hard dependencies should be sequential — the coordinator should maximise concurrency to reduce total execution time. ✓
c. Increase the coordinator's timeout limits to allow more time per subtask.
d. Switch to a faster model for the coordinator.

The correct answer is: Identify which subtasks are independent and dispatch them as parallel Task calls in single coordinator responses. Only tasks with hard dependencies should be sequential — the coordinator should maximise concurrency to reduce total execution time.

Question 7

Flag question

Correct Mark 1.00 out of 1.00

Your research coordinator spawns subagents sequentially: web_search completes, then document_analysis, then fact_checking — three independent tasks that do not depend on each other's outputs. Total latency is 45 seconds. How do you reduce latency?

a. Emit all three Task tool calls in a single coordinator response — the Agent SDK executes multiple Task calls from one response in parallel, reducing total latency to the duration of the slowest single subagent.
b. Run each subagent with a shorter timeout to force faster completion.
c. Upgrade to a faster model to reduce per-subagent processing time.
d. Reduce the number of subagents from three to one with all tools combined.

The correct answer is: Emit all three Task tool calls in a single coordinator response — the Agent SDK executes multiple Task calls from one response in parallel, reducing total latency to the duration of the slowest single subagent.

Question 8

Flag question

Correct Mark 1.00 out of 1.00

Your agent processes financial transactions. The system runs autonomously in production. A developer suggests removing all human-in-the-loop checkpoints to improve throughput. What is the correct response?

a. Replace human review with a second AI agent to maintain oversight without human delays.
b. Remove all checkpoints and monitor for errors post-execution.
c. Remove checkpoints only for low-value transactions to improve throughput.
d. Human-in-the-loop checkpoints should be preserved for irreversible and high-consequence actions regardless of throughput goals. Checkpoints for high-value transactions, account closures, and bulk operations are architectural safety requirements, not optional optimisations.

The correct answer is: Human-in-the-loop checkpoints should be preserved for irreversible and high-consequence actions regardless of throughput goals. Checkpoints for high-value transactions, account closures, and bulk operations are architectural safety requirements, not optional optimisations.

Question 9

Flag question

Correct Mark 1.00 out of 1.00

Your multi-agent pipeline crashes at step 6 of 9. Steps 1–5 wrote results to output files. When you attempt to resume, the coordinator restarts from step 1, ignoring the completed steps. What architectural pattern enables resumption from step 6?

a. Store all intermediate outputs in a database and replay API calls from the beginning.
b. Design each agent to export structured state (completed steps, outputs, status) to a manifest file. On resume, the coordinator reads the manifest, identifies the last completed step, and injects agent states into prompts to continue from step 6. ✓
c. Use --resume to automatically detect and skip completed steps.
d. Always restart from step 1 to ensure state consistency after a crash.

The correct answer is: Design each agent to export structured state (completed steps, outputs, status) to a manifest file. On resume, the coordinator reads the manifest, identifies the last completed step, and injects agent states into prompts to continue from step 6.

Question 10

Flag question

Correct Mark 1.00 out of 1.00

Your agent workflow involves deleting records from a production database at step 4. The agent's test suite passes at 99.7%. During a scheduled run, the agent deletes the wrong records due to a query parameter edge case. What architectural safeguard would have prevented this even if the test suite missed the edge case?

a. Implement a mandatory human-in-the-loop checkpoint before step 4. Irreversible destructive operations on production data require human review regardless of test coverage — test suites cannot guarantee zero edge cases in production. ✓
b. Improve the test suite to cover more edge cases.
c. Run the deletion on a staging environment first as a dry run.
d. Add rollback capability so deleted records can be restored.

The correct answer is: Implement a mandatory human-in-the-loop checkpoint before step 4. Irreversible destructive operations on production data require human review regardless of test coverage — test suites cannot guarantee zero edge cases in production.

Question 11

Flag question

Correct Mark 1.00 out of 1.00

A coordinator's subagent returns: {'status': 'partial', 'completed_topics': ['market size', 'competitors'], 'missing_topics': ['regulatory landscape'], 'reason': 'source unavailable'}. What is the correct coordinator response?

a. Reject the partial result — all topics must be complete before synthesis.
b. Re-run the entire subagent from scratch to get a complete result.
c. Pass the partial result to synthesis without annotation.
d. Use the structured response metadata to make an informed decision: attempt to re-delegate the regulatory landscape to an alternative source subagent, or proceed with synthesis noting the gap explicitly — structured partial results enable intelligent recovery rather than blind success/failure binary decisions. ✓

The correct answer is: Use the structured response metadata to make an informed decision: attempt to re-delegate the regulatory landscape to an alternative source subagent, or proceed with synthesis noting the gap explicitly — structured partial results enable intelligent recovery rather than blind success/failure binary decisions.

Question 12

Flag question

Correct Mark 1.00 out of 1.00

Your agent workflow processes a customer's subscription cancellation. The workflow has 4 steps: verify identity, check cancellation policy, apply cancellation, send confirmation email. After step 3, the cancellation is applied and the subscription is cancelled. The agent then encounters an error on step 4 (email fails to send). How should the agent handle this?

a. Retry step 4 up to 3 times independently — the subscription is already cancelled and that state is correct. If email sending fails after retries, log the failure for human follow-up rather than attempting to undo the cancellation. Each step's state should be treated independently. ✓
b. Start the entire 4-step workflow from the beginning.
c. Roll back step 3 since the workflow did not complete successfully.
d. Escalate to a human for the email failure and also reverse the cancellation.

The correct answer is: Retry step 4 up to 3 times independently — the subscription is already cancelled and that state is correct. If email sending fails after retries, log the failure for human follow-up rather than attempting to undo the cancellation. Each step's state should be treated independently.

Question 13

Flag question

Correct Mark 1.00 out of 1.00

You want to resume a named investigation session from yesterday but your colleague warns that three source files were modified overnight. What is the correct approach to resume safely?

- a. Start a fresh session with a structured summary of yesterday's key findings. Explicitly identify the three modified files in the prompt so the session can re-analyse them with current content. Resuming with stale tool results from modified files produces unreliable analysis. ✓
- b. Resume using --resume and trust the model to detect file changes automatically.
- c. Fork the prior session and re-run only the tool calls on the three modified files.
- d. Resume using --resume but immediately run /compact to clear the stale results.

The correct answer is: Start a fresh session with a structured summary of yesterday's key findings. Explicitly identify the three modified files in the prompt so the session can re-analyse them with current content. Resuming with stale tool results from modified files produces unreliable analysis.

Question 14

Flag question

Correct Mark 1.00 out of 1.00

Your workflow requires a document to be classified before it can be enriched. Enrichment must never run on an unclassified document. With prompt instructions, enrichment runs on unclassified documents in 2% of cases. What achieves 100% enforcement?

- a. Increase the system prompt emphasis: 'NEVER enrich without classifying first.'
- b. Add 50 few-shot examples demonstrating the classify-then-enrich sequence.
- c. Merge classify and enrich into a single combined classify_and_enrich tool.
- d. Implement a programmatic prerequisite gate: a PreToolUse hook that intercepts enrich_document calls and blocks them programmatically unless classify_document has already run and returned a valid classification in the current session. ✓

The correct answer is: Implement a programmatic prerequisite gate: a PreToolUse hook that intercepts enrich_document calls and blocks them programmatically unless classify_document has already run and returned a valid classification in the current session.

Question 15

Flag question

Correct Mark 1.00 out of 1.00

A new developer joins the team and asks: 'Why do we check stop_reason == end_turn to terminate the loop instead of checking if the response has no tool_use blocks?' What is the correct explanation?

a. A response can contain both text blocks AND tool_use blocks simultaneously. Checking for the absence of tool_use blocks would terminate the loop even when the model intends to make a tool call alongside explanatory text. Only stop_reason provides a definitive signal that the model considers the task complete. ✓

b. Both methods are acceptable depending on the use case.

c. Checking for no tool_use blocks is equivalent and equally reliable.

d. stop_reason is a legacy field — checking tool_use blocks is the modern approach.

The correct answer is: A response can contain both text blocks AND tool_use blocks simultaneously. Checking for the absence of tool_use blocks would terminate the loop even when the model intends to make a tool call alongside explanatory text. Only stop_reason provides a definitive signal that the model considers the task complete.

Question 16

Flag question

Correct Mark 1.00 out of 1.00

Your subagent successfully completes its assigned research on 'AI investment trends in Southeast Asia' but the coordinator has no way to tell which specific countries were covered versus which were missed. How should the subagent's output be structured to provide this transparency?

a. The coordinator should ask the subagent for a coverage list in a follow-up Task.

b. Include explicit coverage metadata in the subagent's structured output: a list of entities covered, a list of entities not covered due to data availability, and a coverage completeness flag. This enables the coordinator to identify gaps and decide whether to re-delegate. ✓

c. Return only the findings — the coordinator can infer coverage from the content.

d. Coverage transparency is the synthesis agent's responsibility, not the subagent's.

The correct answer is: Include explicit coverage metadata in the subagent's structured output: a list of entities covered, a list of entities not covered due to data availability, and a coverage completeness flag. This enables the coordinator to identify gaps and decide whether to re-delegate.

Question 17

Flag question

Correct Mark 1.00 out of 1.00

Your coordinator's task decomposition generates subtasks like: 'research AI in healthcare', 'research AI in healthcare applications', and 'find AI healthcare use cases'. All three overlap heavily. What is the impact and fix?

a. The coordinator should merge all three into one subtask to save resources.
b. Redundant subtasks waste compute, produce duplicate findings, and make synthesis difficult. Decompose with distinct, non-overlapping scope assignments — by sector, geography, time period, or data type — ensuring collective coverage without duplication. ✓
c. Overlapping subtasks improve coverage through redundancy — this is acceptable.
d. Overlapping subtasks are only a problem when more than 5 subagents are used.

The correct answer is: Redundant subtasks waste compute, produce duplicate findings, and make synthesis difficult. Decompose with distinct, non-overlapping scope assignments — by sector, geography, time period, or data type — ensuring collective coverage without duplication.

Question 18

Flag question

Correct Mark 1.00 out of 1.00

Your subagent system prompt says 'use all available tools to complete your research task.' You observe the synthesis subagent performing web searches even though it should only synthesise findings already passed to it. What is the correct design change?

a. Remove the synthesis subagent's system prompt entirely.
b. Give the synthesis subagent a PreToolUse hook that blocks web search calls.
c. Apply principle of least privilege: restrict the synthesis subagent's allowed_tools to only what it needs for synthesis — typically no search tools — preventing cross-specialisation misuse. ✓
d. Add a system prompt instruction: 'Do not use search tools unless necessary.'

The correct answer is: Apply principle of least privilege: restrict the synthesis subagent's allowed_tools to only what it needs for synthesis — typically no search tools — preventing cross-specialisation misuse.

Question 19

Flag question

Correct Mark 1.00 out of 1.00

Your coordinator agent's context window is growing large after processing many subagent results. The developer suggests keeping full verbatim subagent outputs in the coordinator's context for maximum information retention. What is the correct approach?

a. Use a separate long-context model for the coordinator to handle large outputs.
b. Delete all subagent outputs after synthesis to free context space.
c. Compress each subagent's verbose output into a structured summary containing key claims, source references, and confidence scores before aggregating in the coordinator's context. This prevents context overflow while preserving the information needed for synthesis. ✓
d. Always keep full verbatim outputs — information loss in summaries creates errors.

The correct answer is: Compress each subagent's verbose output into a structured summary containing key claims, source references, and confidence scores before aggregating in the coordinator's context. This prevents context overflow while preserving the information needed for synthesis.

Question 20

Flag question

Correct Mark 1.00 out of 1.00

Your coordinator agent produces excellent results on simple queries but makes poor decisions on complex multi-faceted requests. Investigation shows it always routes complex requests through the same subagent pipeline regardless of the specific complexity dimensions. What should the coordinator do differently?

a. Route all complex requests to the most capable (highest-tier) subagent.
b. Use a longer timeout for complex requests to allow more processing time.
c. Apply adaptive task decomposition: analyse the specific dimensions of complexity in each request and generate subtasks tailored to those dimensions, invoking different specialist subagents based on what the query actually requires rather than applying a fixed pipeline to all complex inputs. ✓
d. Add more subagents so complex requests have more paths available.

The correct answer is: Apply adaptive task decomposition: analyse the specific dimensions of complexity in each request and generate subtasks tailored to those dimensions, invoking different specialist subagents based on what the query actually requires rather than applying a fixed pipeline to all complex inputs.

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